

Lab 2

5% Overall Grade

This lab requires you to download a folder of files from GitHub onto your local computer, upload that folder to an AWS S3 bucket, and explore various file access methods. You will compare how files are accessed via the S3 object URL versus a website URL (which will fail if the bucket is private), configure a lifecycle policy, and generate pre-signed URLs for secure sharing.

Part A: Download Files from GitHub

1. Download the provided folder from the GitHub repository onto your local computer
 - https://github.com/SohailHanif/Bioinformatics_Cloud_Computing
2. Verify the folder contains the lab_2 folder
3. Screenshot Requirements:
 - Folder contents visible on your local computer

Part B: Upload Folder to AWS S3

1. Log in to the AWS Management Console and navigate to the S3 service
2. Create a new S3 bucket with a name that includes your first name and lab identifier (e.g., FIRSTNAME_Lab_2_Bucket)
3. Upload the entire folder you downloaded from GitHub to the S3 bucket
4. Screenshot Requirements:
 - S3 bucket creation confirmation
 - Uploaded folder with visible file structure in S3

Part C: Viewing Files in S3

1. In the S3 Console, select the fast_qc_report.html, snpEff_annotation.html and click the Open button; note that it should render in your browser
2. Copy the Object URL and paste it into a new browser tab to confirm it renders correctly
3. Ensure the bucket's public access settings are set to private and then attempt to access the file using the website URL; observe that access fails
4. Screenshot Requirements:
 - Browser view of index.html rendered via S3 Object URL
 - Error message when attempting to access via website URL

Part D: Configure Lifecycle Policies

1. In your S3 bucket, navigate to the Management tab and create a new Lifecycle Rule
2. Configure the rule to transition objects to a lower-cost storage class (e.g., S3 Infrequent Access) after 90 days

3. Screenshot Requirements:

- Lifecycle rule configuration screen
- Active lifecycle rule list in the S3 bucket

Part E: Generate and Test Pre-Signed URLs

1. In the S3 Console, select one of the uploaded files and generate a pre-signed URL with an expiration time (e.g., 1 hour)
2. Copy the pre-signed URL and open it in an incognito browser window to verify that the file is accessible without credentials
3. Screenshot Requirements:
 - Pre-signed URL generation process
 - Browser view showing successful access via the pre-signed URL

Files to Submit:

- A PDF summarizing your process and observations
- Screenshots of:
 - Local folder contents from the GitHub download
 - S3 bucket creation and folder upload
 - Browser view of file rendering via S3 Object URL
 - Error message when accessing file via website URL
 - Lifecycle rule configuration and active rule list
 - Pre-signed URL generation and successful access via incognito window